

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

Unit 15: Context Awareness

1

Cheng-Hsin Hsu (chsu@cs.nthu.edu.tw)

Some slide adopted from Prof. Christian Poellabauer's materials

Agenda

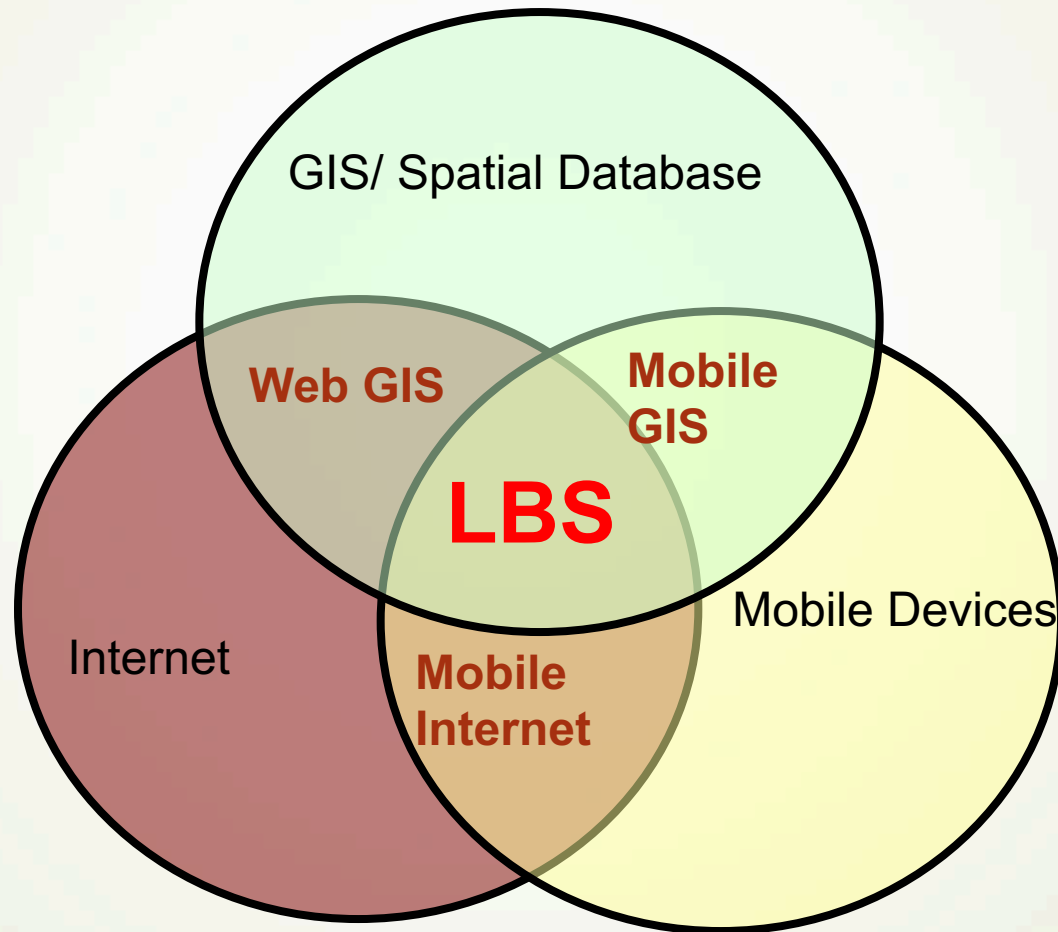
- Location-Based Services
- Context-Aware Systems
- Challenges and Design Guidelines
- Sample Work: Coordinated Sensor Activations

Location-based Services: Definition

LBS: A certain *service* that is offered to the users *based* on their *locations*.



Convergence of Technologies



Origin of Location-Based Services

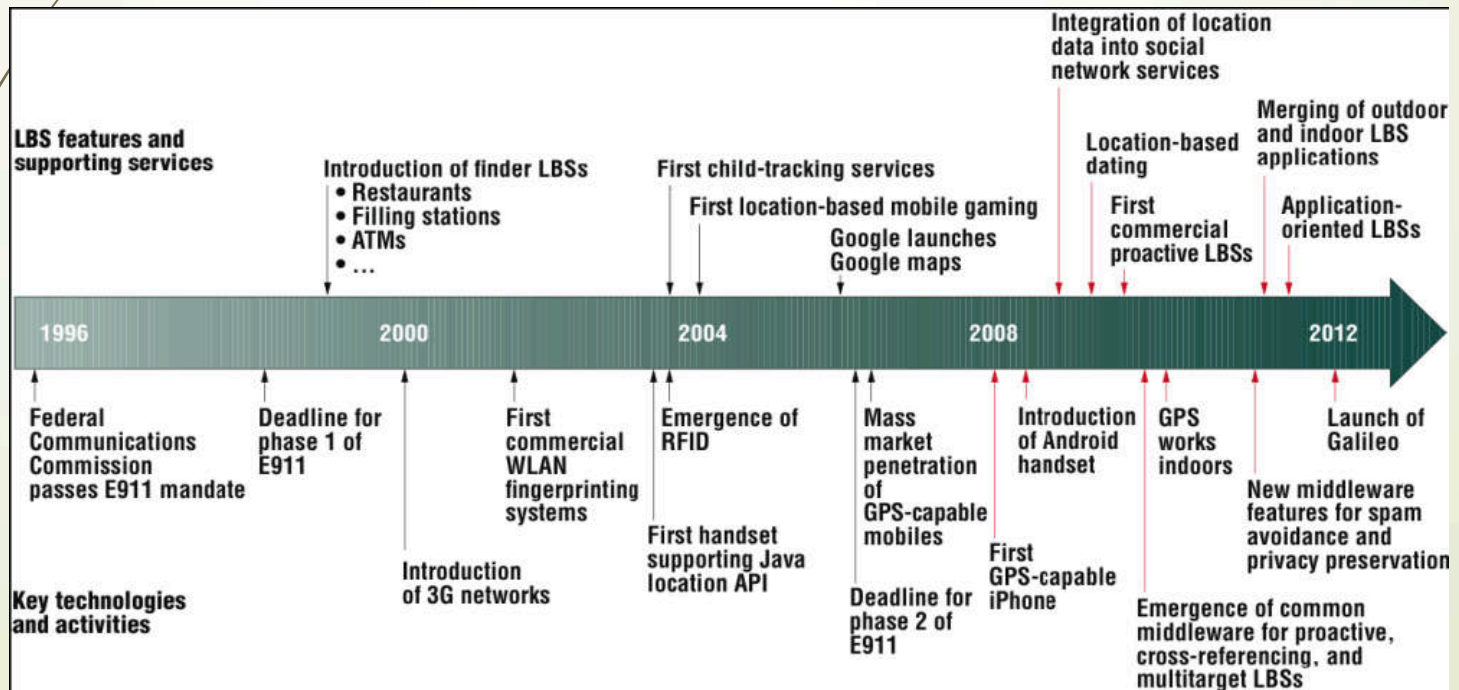
- The main origin of Location-Based Services (LBS) was the **E911** (Enhanced 911) mandate, which the U.S. government passed in 1996.
- The mandate was for mobile-network operators to locate **emergency callers** with prescribed accuracy, so that the operators could deliver a caller's location to Public Safety Answering Points.
- Cellular technology couldn't fulfill these accuracy demands back then, so operators started enormous efforts to introduce advanced positioning methods.

(The Failed) Finder Services

- To gain returns on the E911 investments, operators launched a series of commercial LBSs.
- In most cases, these consisted of **finder services** that, on request, delivered to users a list of nearby points of interest, such as restaurants or gas stations.
- However, most users weren't interested in this kind of LBS, so many operators quickly phased out their LBS offerings and stopped related development efforts.
- **Limitations: Reactive, self-referenced, content-oriented, operator-centric**

Situations Changed as the Technologies Advance

- The emergence of **GPS**-capable mobile devices, the advent of the Web 2.0 paradigm, and the introduction of 3G broadband wireless services were among the enabling developments.



New (and Successful) Location-Based Services

- In 2004, operators and other providers started offering services for **fleet management** and for **tracking children and pets**—these were the first examples of **cross-referencing LBSs**.
- Initial versions of these services were based on **cell-ID positioning** using triangulation techniques, which suffered from low accuracy and were soon replaced by **GPS**.
- An **overlay of geo-location technologies** consisting of cellular and Wi-Fi triangulations, in addition to low-power GPS receivers (e.g., assisted GPS), made it possible for location information to be available most of the time and with variable accuracies.

Maps for Location-Based Services

- **Interactive digital** maps; used in many applications, with many map features (location, navigation, nearby sites, traffic overlay, ...)
- The world of digital navigable maps can be traced back to **NAVTEQ**, the most dominant company in geographic information systems and electronic maps.
- **Indoor** maps, e.g., major airports, shopping malls, stadiums, resorts and other complex architectural spaces; seamlessly embedded and laid over outdoor maps, which requires no switching actions by the users - only zooming is required to see the details of an indoor map.
- Both iOS and Android provide comprehensive APIs.

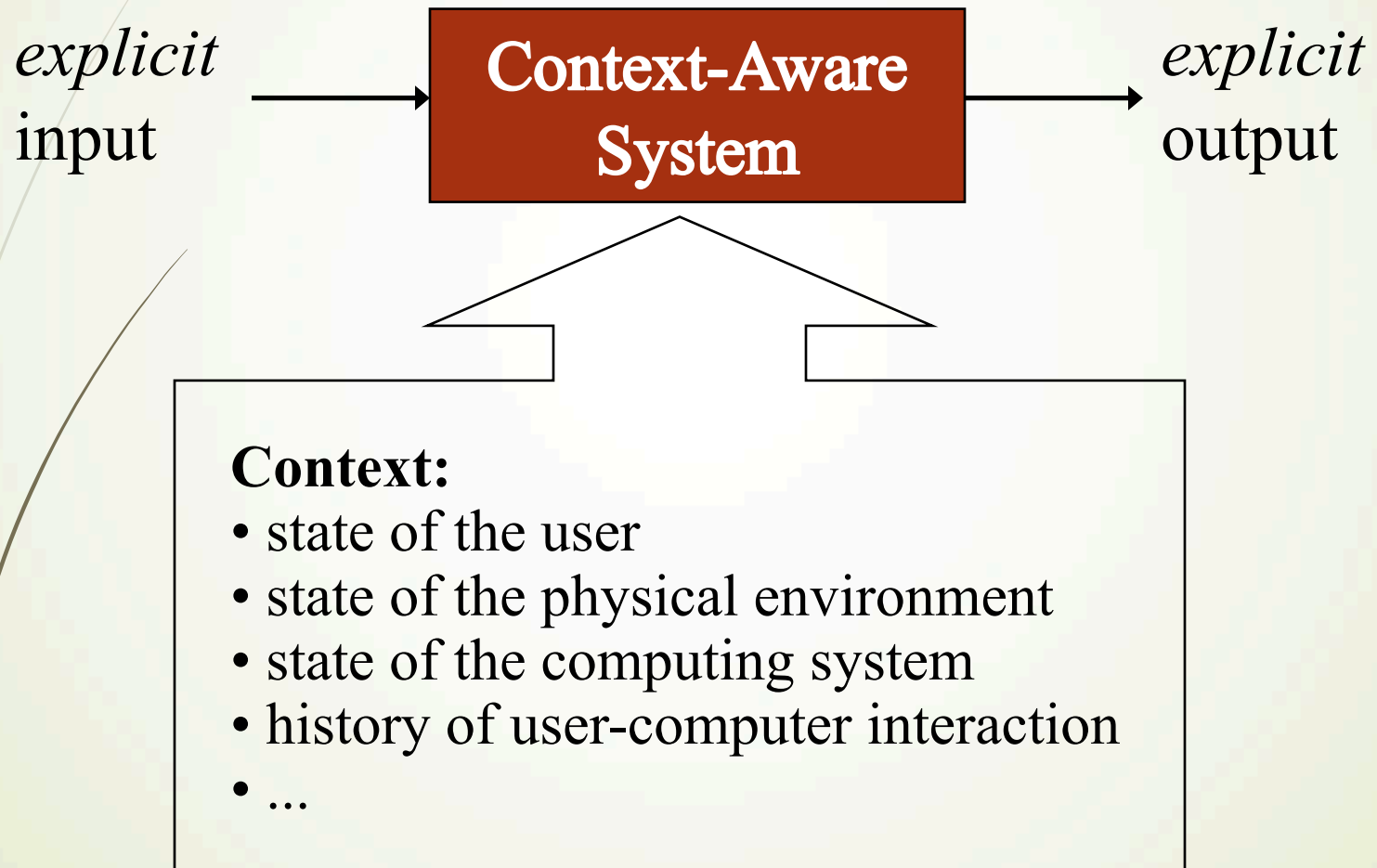
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System Structure (Without Context Awareness)



Context as **Implicit** Input



What is Context?



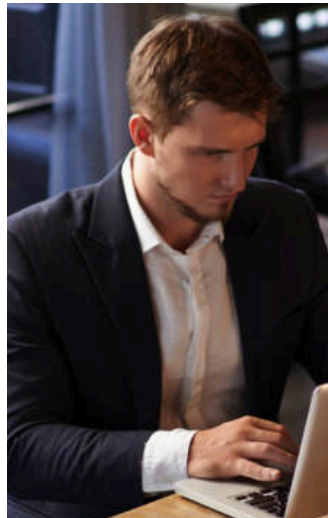
Examples of Context

- Identity (user, others, objects)
- Location
- Date/Time
- Environment
- Emotional state
- Focus of attention
- Orientation
- User preferences
- Calendar (events)
- Browsing history
- Behavioral patterns
- Relationships (phonebook, call history)
- ... the elements of the user's environment that the computer knows about...

Relevance of Context Information

- Trying to arrange lunch meeting
- Going to a job interview
- Going home after work and making evening plans
- Shopping
- Tourist
- ...

Sample Scene 1



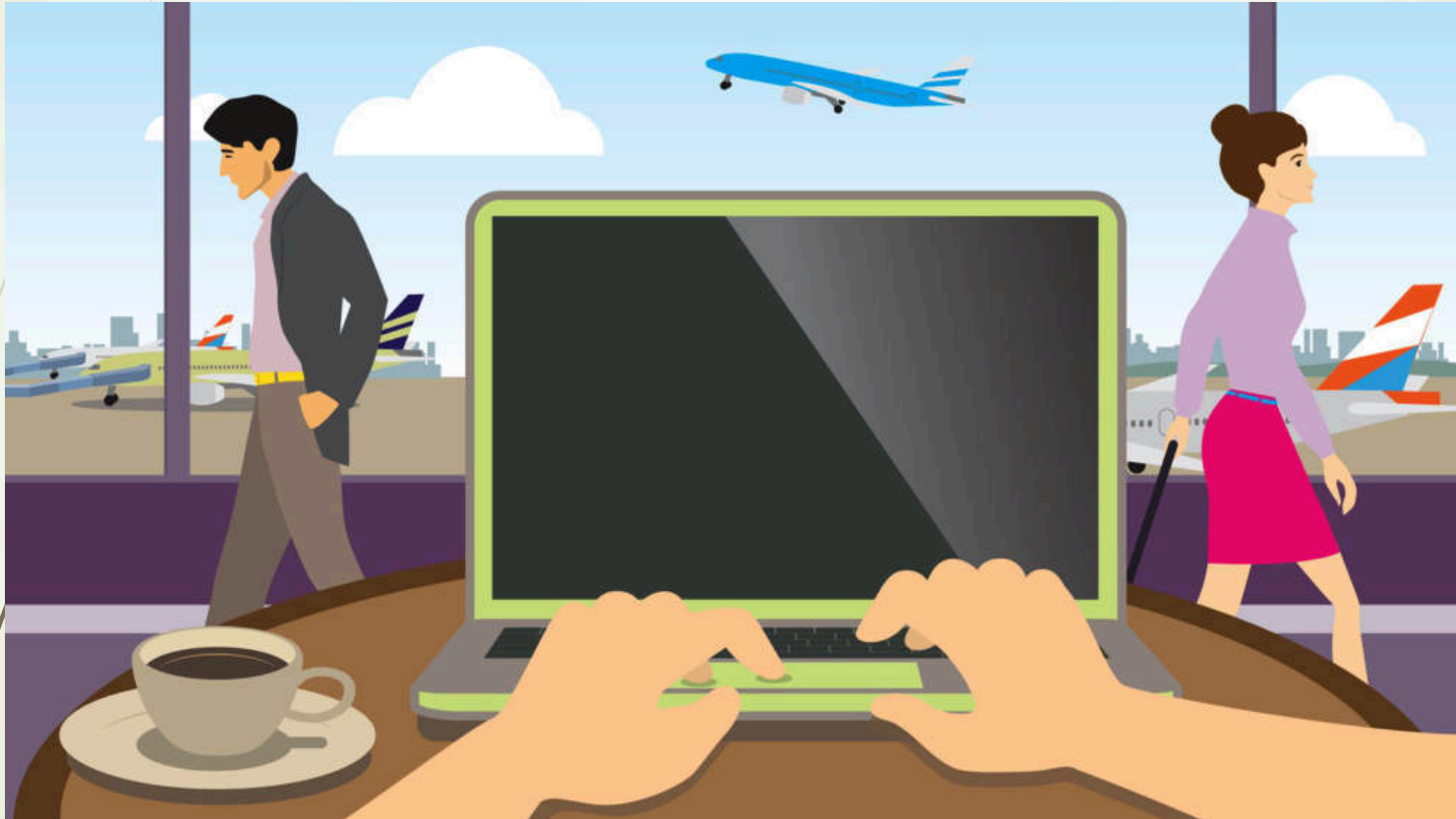
Sample Scene 1



Sample Scene 2



Sample Scene 2



Examples

- Smartphone adjusts the screen to the orientation of the device
- Apple Watch turns on display if arm lifted/rotated
- Orientation is determined by using both a gyroscope and an accelerometer

